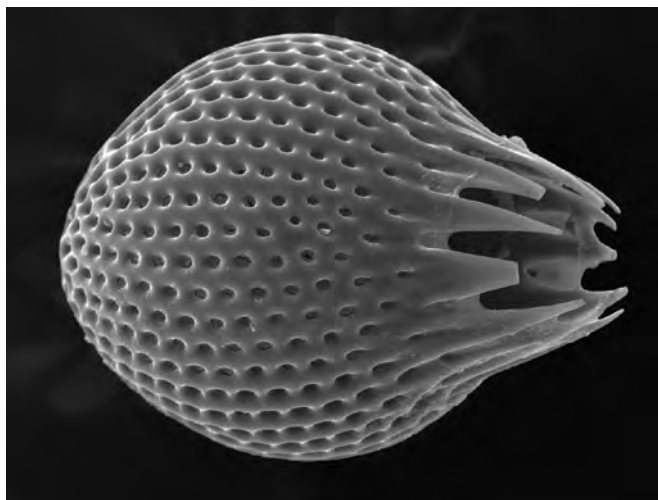


2 Diatoms are single-celled organisms important for many ocean food chains.

The photograph shows a diatom, as seen using a scanning electron microscope.



(Source: © AMI IMAGES / SCIENCE PHOTO LIBRARY)

(a) Diatoms contain a nucleus, mitochondria and chloroplasts.

(i) State the domain which contains diatoms.

(1)

(ii) The nucleus in a diatom contains DNA.

Name **two processes** that take place in the nucleus of a diatom.

(2)

1 .....

.....

.....

2 .....

.....

.....



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- (b) It is predicted that diatom populations will decrease in number due to global warming.

Scientists have suggested that this will cause a decrease in ocean biodiversity.

Describe how a **decrease** in biodiversity within an **ocean** habitat could be determined.

(3)

(Total for Question 2 = 6 marks)

